

LAB-4

Aim : To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

Q1. Find match Ids of all the matches in the database in which tendulkar batted.

```
mysql> SELECT MatchID FROM Batting, Player;
```

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;
```

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND  
Lname='Tendulkar';
```

OR

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM  
Player WHERE LName='Tendulkar'); +-----+
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;
+-----+
| MatchID |
+-----+
| 2755 |
| 2755 |
| 2755 |
| 2755 |
| 2755 |
| 2689 |
| 2689 |
| 2689 |
| 2689 |
| 2689 |
| 2689 |
| 2689 |
| 2689 |
| 2755 |
| 2689 |
| 2755 |
+-----+
16 rows in set (0.0017 sec)

MySQL localhost:3306 ssl lakshay_match SQL SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0035 sec)
```

Q2. Find match information of all matches in which Dohni has batted.

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;
```

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN
(SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player
WHERE LName ='Dhoni'));
```

OR

Using Join

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE
ODIMatch.MatchID=Batting.MatchID AND PlayerID=PID AND LName='Dhoni';
```

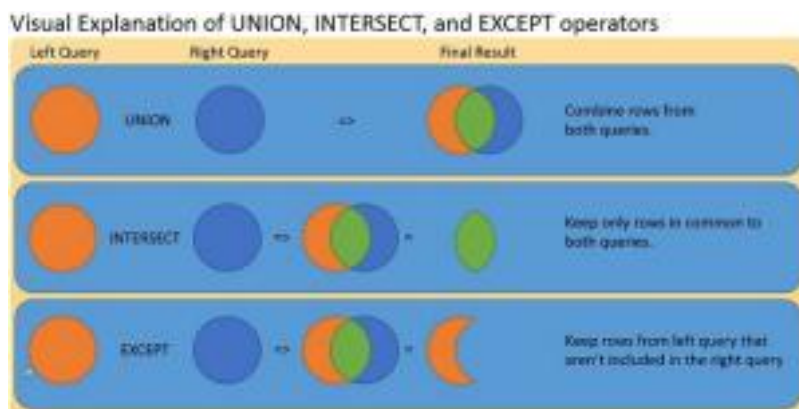
```
MySQL localhost:3306 ssl lakshay_match SQL SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE ODIMatch.MatchID=Batting.MatchID AND Play
erID=PID AND LName='Dhoni';
+-----+-----+-----+-----+
| Team1 | Team2 | Ground | Date |
+-----+-----+-----+-----+
| Sri Lanka | India | Colombo | 2008-08-24 |
| Australia | India | Brisbane | 2008-03-04 |
+-----+-----+-----+-----+
2 rows in set (0.0076 sec)
```

Q3. Find match Ids of all matches in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM
Player WHERE Lname='Tendulkar'); OR
```

Using Join

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND
LName='Tendulkar';
```



```
MySQL localhost:3306 ssl lakshay_match SQL SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0037 sec)
```

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID
FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689)
+----+
| PID |
+----+
| 23001 |
| 24001 |
| 99001 |
| 98002 |
+----+
```

Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID
FROM Bowling WHERE MatchID=2689);
```

Using Union

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID
FROM Bowling WHERE MatchID=2689); +-----+
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 89001 |
| 23001 |
| 25001 |
| 99001 |
| 95002 |
| 95001 |
| 24001 |
| 99001 |
| 27001 |
| 98002 |
+----+
9 rows in set (0.0013 sec)
```

Q6. Find Ids

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT
PID FROM Bowling WHERE MatchID=2689); +-----+
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 89001 |
| 25001 |
| 99002 |
| 95001 |
| 27001 |
+----+
5 rows in set (0.0013 sec)
```

Q7. Find match Ids where Tendulkar has played.

```
mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE
LName='Tendulkar') ST WHERE PID=ST.PlayerID; +-----+
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0018 sec)
```

Q8.Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL
(SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN
(SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND
MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0048 sec)
```

Q8.Find match Ids of all those matches in which player 27001 has scored minimum runs than he made at every match he played at Brisbane

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL
(SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN
(SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND
MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
+-----+
| MatchID |
+-----+
| 2755 |
+-----+
1 row in set (0.0013 sec)
```

Q9(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND
MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WH
ERE Ground='Colombo');
+-----+-----+
| PID | NRuns |
+-----+-----+
| 25001 | 71 |
| 91001 | 60 |
+-----+-----+
2 rows in set (0.0009 sec)
```

Q9(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN
(SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'); +-----+-----+
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
+-----+
| PID | NRuns |
+-----+
| 25001 | 71 |
| 91001 | 60 |
+-----+
2 rows in set (0.0013 sec)
```

Q10. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

```
mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PlayerID FROM Player WHERE FTest IS NULL;
+-----+
| PlayerID |
+-----+
| 27001 |
+-----+
1 row in set (0.0013 sec)
```

Q11. Find names of the player who batted in match 2689

```
mysql> SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);
```

OR

```
mysql> SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND MatchID=2689;
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);
+-----+
| FName | LName |
+-----+
| Sachin | Tendulkar |
| Yuvraj | Singh |
| MS | Dhoni |
| Adam | Gilchrist |
| Ricky | Ponting |
| Andrew | Symonds |
| Breet | Lee |
| Praveen | Kumar |
| Harbhajan | Singh |
+-----+
9 rows in set (0.0015 sec)
```

Q12. Find player Ids of player who have scored more than 30 in every ODI match that they have played.

```
mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31); +-----+
```

+

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND N
Runs < 31);
+-----+
| PlayerID |
+-----+
| 24001 |
| 25001 |
| 25001 |
| 89001 |
| 91001 |
+-----+
5 rows in set (0.0012 sec)
```

Q13. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

SELECT FName, LName

FROM Player

WHERE NOT EXISTS

(SELECT *

FROM ODIMatch

WHERE Ground = 'Melbourne'

AND NOT EXISTS

(SELECT *

FROM Batting

WHERE PID=PlayerID

AND (Batting.MatchId=ODIMatch.MatchID));

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT FName, LName FROM Player WHERE NOT EXISTS ( SELECT * FROM ODIMatch WHERE Ground = 'Melbo
urne' AND NOT EXISTS ( SELECT * FROM Batting WHERE PID = PlayerID AND Batting.MatchID = ODIMatch.MatchID
));
Empty set (0.0016 sec)
```

Q14. Find the number of players that bowled in ODI match 2689

mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
+-----+
| NBowlers |
+-----+
| 6 |
+-----+
1 row in set (0.0038 sec)
```

Q15. Find the average batting score of all the players that batted in the ODI Match 2689

```
mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
+-----+
| AvgRuns_2689 |
+-----+
|      25.2222 |
+-----+
1 row in set (0.0043 sec)
```

Q16. Find the youngest player in the database

```
mysql> SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT
MAX(YBorn) FROM Player);
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
+-----+
| Youngest_Player |
+-----+
| Ishant          |
+-----+
1 row in set (0.0040 sec)
```

Q17. Find the number of players in the database from each country

```
mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
+-----+-----+
| Country | NPlayers |
+-----+-----+
| India   | 10       |
| Australia | 5       |
| West Indies | 1       |
| Sri Lanka | 3       |
| Pakistan | 2       |
| Zimbabwe | 1       |
| South Africa | 4       |
| New Zealand | 2       |
| England | 1       |
+-----+-----+
9 rows in set (0.0010 sec)
```

Q18. Find batting average of each player in the database. Present the PID of each Player.

```
mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

PID	AVGR
23001	19.0000
24001	42.0000
25001	53.5000
27001	4.5000
89001	91.0000
91001	60.0000
92002	1.0000
94002	17.0000
95001	1.0000
98002	2.5000
99001	7.0000
99002	2.0000

12 rows in set (0.0010 sec)

Solve the following

Q19. Find batting average of each player from India in the database. Present first name of each player.

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT P.FName, AVG(B.NRuns) AS Batting_Average FROM Player P JOIN Batting B ON P.PlayerID = B.PID WHERE P.Country = 'India' GROUP BY P.PlayerID, P.FName;
```

FName	Batting_Average
Yuvraj	19.0000
MS	53.5000
Praveen	4.5000
Sachin	91.0000
Harbhajan	2.5000

5 rows in set (0.0012 sec)

Q20. Find the average score for each player when playing against Australia

```
MySQL localhost:3306 ssl lakshay_match SQL SELECT P.FName, P.PlayerID, AVG(B.NRuns) AS AvgScoreVsAustralia FROM Player P JOIN Batting B ON P.PlayerID = B.PID JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Australia' OR M.Team2 = 'Australia' GROUP BY P.PlayerID, P.FName;
```

FName	PlayerID	AvgScoreVsAustralia
Sachin	89001	91.0000
Yuvraj	23001	38.0000
MS	25001	36.0000
Adam	99002	2.0000
Ricky	95001	1.0000
Andrew	24001	42.0000
Breet	99001	7.0000
Praveen	27001	7.0000
Harbhajan	98002	3.0000

9 rows in set (0.0016 sec)

Q21. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

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```
MySQL localhost:3306 ssl lakshay_match SQL SELECT B.PID, AVG(B.NRuns) AS Average_Score FROM Batting B JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M
.Team1 = 'Sri Lanka' OR M.Team2 = 'Sri Lanka' GROUP BY B.PID ORDER BY Average_Score DESC;
+-----+-----+
| PID | Average_Score |
+-----+-----+
| 25001 | 71.0000 |
| 91001 | 60.0000 |
| 94002 | 17.0000 |
| 27001 | 2.0000 |
| 98002 | 2.0000 |
| 92002 | 1.0000 |
| 23001 | 0.0000 |
+-----+-----+
7 rows in set (0.0013 sec)
```